BASIC SCIENCE, YEAR 7

2024/2025 ACADEMIC SESSION

SECOND TERM LESSON NOTES

SCHEME OF WORK

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Week One

ENVIRONMENTAL POLLUTION

Lesson Objectives

By the end of this lesson, students should be able to:

(a) define environmental pollution
(b) state the causes of air and water pollution.
(c) mention the effects of each of air and water pollution.
(d) explain the causes and effects of each of air and water pollution.
(e) discuss ways to control air and water pollution.
(f) itemize techniques of purifying polluted water.

REFERENCE TEXT: Classic Basic Science for Junior Secondary Schools, Book 1, Chapter 2, Pages 25 -32

Pollution

This occurs when harmful substances are released into the environment. These harmful substances are known as POLLUTANTS.

Types of Environmental Pollution

1. Air Pollution

Air pollution occurs when harmful substances are released into the atmosphere as a result of human activities such as bush burning, burning of fuel, use of chemicals, industrialization etc.

Air pollutants include;

1. Lead dust
2. smoke and soot
3. Carbon monoxide
4. Sulphur dioxide
5. Nitrogen dioxide
6. Chlorofluorocarbons (CFC's)
7. Dust particles

Causes Of Air Pollution

1. Burning of firewood, coal, refuse, petroleum products.
2. Road construction.
3. Sweeping of dusty floors
4. Chemicals such as insecticides, perfumes, coolants in air conditioners and refrigerators.
5. Quarries (places where heavy stones and rocks are crushed)

1. Sewage should be treated and converted to manure.
2. Refuse should be buried or burnt in an incinerator.
3. Chemicals should be converted to harmless substances.
4. Crude oil spills should be prevented.
5. Excess use of chemicals should be stopped.

Week Two

4. Noise Pollution

This is when noise is introduced into the environment. Noise is a disorderly sound that can cause harm to man and animals. Examples of noise pollutants includes;

(a) market noise
(b) public motor parks
(c) generators
(d) trains
(e) aeroplanes
(f) building construction
(g) vehicle horns
(h) church/mosque vigil
(i) public address system

Causes Of Noise Pollution

1. Different loud conversations at public car parks and market places.
2. The use of generators to provide electricity.
3. The use of loud vehicle horns.
4. The use of heavy duty factory machines.
5. Loud music, prayers and preaching in churches and mosques.
6. Moving airplanes and trains.
7. Loud public address system.

Effects Of Noise Pollution

1. Hearing loss
2. Discomfort
3. Inability to concentrate
4. Hypertension
5. Sleep disturbance
6. Accident

Control Of Noise Pollution

1. Seperating industrial areas from residential areas.
2. The use of sound-proof generators.
3. Avoid unnecessary blasting of car horns.
4. Locating public motor parks far from residential areas.
5. Laws should be put in place to prevent street parties where public address system can cause noise pollution.

SEXUALLY TRANSMITTED INFECTIONS (STI)

Lesson Objectives

By the end of this lesson, students should be able to:

(a) define this sexually transmitted infections
(b) list ways STI's can be spread.
(c) itemize some STI's and give their symptoms.
(d) explain the effects of STI's.
(e) itemize ways to prevent the spread of STI's.

REFERENCE TEXT: Classic Basic Science for Junior Secondary Schools, Book 1, Chapter 2, Pages 13 -22

Sexually Transmitted Infections (STI)

These are infections that can be gotten when a person has sexual contact with another person that is infected.

Ways STI can be spread include;

1. By sharing needles or using unsterilized needles for body piercing and tattooing the body.
2. Sexual contact with an infected person.
3. From a pregnant mother to her unborn baby.
4. From a breastfeeding mother to her nursing baby.

Examples of STI

1. *Gonorrhea:* It is transmitted during sexual intercourse and during childbirth from an infected mother. Its symptoms include;
    a. Pains when urinating
    b. Yellow discharge from the urinary organs.
2. *Syphilis:* It is transmitted during sexual intercourse and from direct contact with a syphilis sore on the anus, vagina or mouth. It can also be transmitted from a pregnant mother to her unborn baby. Its symptoms include;
    a. Small painless sores for the first 3-6 weeks.
    b. Skin rashes on the palms and soles of the feet.
    c. Sore throat, muscle aches and headaches.
    d. Damage of the internal organs which can lead to death.
3. Chlamydia: It is transmitted during sexual intercourse. Its symptoms include;
    a. Abnormal discharge from the vagina or penis.
    b. Burning feeling when urinating.
    c. Itching around the opening of the penis.
    d. Bleeding between menstruations.
    e. Pains during intercourse.

HIV/AIDS

Lesson Objectives

By the end of this lesson, students should be able to:

(a) give the full meaning of HIV and AIDS.
(b) differentiate between HIV and AIDS.
(c) mention the modes of transmission of HIV.
(d) discuss the behaviours that put people at risk of HIV.
(e) Itemize ways to prevent the spread of HIV.

REFERENCE TEXT: Classic Basic Science for Junior Secondary Schools, Book 1, Chapter 2, Pages 13 -22

HIV/AIDS

HIV means Human Immunodeficiency Virus. It is caused by a virus that destroys the white blood cells in the body that help us to fight infections and diseases. When this happens, the body's immune system is destroyed.

AIDS means Acquired Immune Deficiency Syndrome and it develops once the body's immune system has been weakened by HIV. At this stage, the body can no longer fight infections.

Difference between HIV and AIDS

HIV is the virus that causes AIDS. A person with AIDS has no immunity to any disease he/she comes in contact with and at this stage, it can lead to death.

Mode Of Transmission

1. Sexual intercourse without using condom.
2. Blood transfusion from an infected person.
3. Using infected and unsterilized sharp objects like blades, knives, needles etc.
4. From an infected pregnant woman to her unborn baby.
5. By sharing toothbrushes.

Behaviours That Put People at Risk

1. Engaging in unprotected sex.
2. Assuming one's partner is disease free based on appearance.
3. Having multiple sex partners.
4. Sharing unsterilized syringe and needles.
5. Using drugs and/or alcohol which makes it harder to abstain from sex or practice safe sex.

Week Four

HIV/AIDS II

Lesson Objectives

By the end of this lesson, students should be able to:

(a) discuss the purpose of testing and counselling for HIV.
(b) describe some HIV counselling centers.
(c) explain ways to help and support people with HIV
(d) itemize some myths about HIV.

REFERENCE TEXT: Classic Basic Science for Junior Secondary Schools, Book 1, Chapter 2, Pages 13 -22

Counselling

Counselling is a private conversation with a trained person with the aim of helping you understand and help yourself.

Counselling and testing is when a person chooses to receive counselling on HIV/AIDS to allow them take decisions regarding their HIV status. HIV/AIDS counselling is free and confidential. HIV/AIDS counselling centers include;

- sexual health clinics

- health centers

- teaching hospitals

- general hospitals.

HIV/AIDS has no cure but it can be controlled. It is very important to undergo HIV testing at regular intervals to know your status.

Care and Support

People living with HIV/AIDS need care and support from their family, friends and the community especially when they are ill. Friends and family members should understand that HIV/AIDS cannot be spread by coughing, touching or sharing eating utensils.

We can help them by doing the following;

a. Showing them love, respect and support.
b. Helping them to reduce stress.
c. Providing them with balanced and nutritious meals.
d. Talking openly with them about the disease knowing the facts about HIV/AIDS.
5. Encouraging them to get treatments if they are sick.

Week Five

ENERGY

Lesson Objectives

By the end of this lesson, students should be able to:

(a) define energy.
(b) list the sources of energy and give examples of each.
(c) Itemize and the sources of energy
(d) State the law of conversion/conservation of energy.
(e) explain the transformation of energy

REFERENCE TEXT: Classic Basic Science for Junior Secondary Schools, Book 1, Chapter 2, Pages 13 -22

Energy

Energy is defined as the ability to do work. When something can do work it is said to possess energy.

Sources of energy

There are two sources of energy;

a. Natural sources of energy
b. Artificial sources of energy

1. Natural sources of energy: These are energy sources that are not made by man. Examples of natural sources of energy are;

a. Food: We obtain chemical energy from food we eat. Without food we would die.
b. The sun: Light and heat energy are gotten from the sun. The sun is a universal source of energy and humans depends on energy from the sun to survive.
c. Coal: Coal is gotten from the remains of dead plant materials that have undergone intense heat and pressure for millions of years. Energy is produced when coal is burnt in air.
d. Natural gas: Natural gas is found with petroleum under the earth. It is burnt to produce heat energy which is used for cooking and to produce electricity.
e. Wood: Wood burns to give fire. It is a source of heat energy.
f. Water: Moving water such as rivers and waterfalls are used to move turbines in a dam. The turbines in turn generated energy which is used to power an electric generator.
g. Wind: Moving air is known as wind. Wind energy is used to produce electricity by turning the turbines of a windmill.

2. Artificial sources of energy: These are energy sources made by man.

a. Batteries: Batteries are used to store chemical energy. They can be used to produce electricity. Primary batteries cannot be recharged e.g. torch batteries while secondary batteries can be recharged.

Telephones and radios work through energy conversion. A telephone has one end for speaking and another for receiving. When you speak into the mouth piece end, sound energy is converted into electrical energy.

A radio works in a similar way.

4. Mechanical energy to heat energy: This energy conversion occurs when striking a match, rubbing your palms together, Sharpening a knife on a hard surface etc.
5. Solar energy to electrical energy e.g. Solar panels.
6. Potential energy to kinetic energy: This occurs when an object is dropped from a tall building, when a ball is thrown into the air, when a rubber band is stretched etc.
7. Electrical to heat energy: This is common in household appliances like electric kettles, cookers, boiling rings etc.

16	Sulphur	S
17	Chlorine	Cl
18	Argon	Ar
19	Potassium	K
20	Calcium	Ca

The first 20 elements are very important because they are found in common chemical processes and cut across the various groups on the periodic table.

Symbols of Elements

Symbols of elements consists of one or two letter abbreviation of the elements name. There are 4 ways symbols of elements are derived;

1. From the first letter of the elements name.

Elements	Symbol
1. Hydrogen	H
2. Oxygen	O
3. Nitrogen	N
4. Sulphur	S
5. Carbon	C
6. Iodine	I
7. Phosphorus	P

2. From the first letter and the second letter in the name of the element.

Elements	Symbol
1. Beryllium	Be
2. Helium	He
3. Neon	Ne
4. Aluminium	Al
5. Silicon	Si
6. Calcium	Ca

Compound	**Combining elements**	**Chemical formula**
1. Water	Hydrogen and oxygen	H_2O
2. Table salt (sodium chloride)	Sodium and chlorine	NaCl
3. Hydrogen peroxide	Hydrogen and oxygen	H_2O_2
4. Ammonia	Nitrogen and hydrogen	NH_3
5. Hydrochloric acid	Hydrogen and chlorine	HCl
6. Carbon dioxide	Carbon and oxygen	CO_2
7. Sodium bicarbonate (baking soda)	Sodium, hydrogen, carbon, oxygen.	$NaHCO_3$
8. Sucrose (sugar)	Carbon, Hydrogen, Oxygen	$C_{12}H_{12}O_{11}$

7. Trace Gases such as Helium, Lithium, Methane etc. in very small quantity.

Usefulness Of Air Components

1. Nitrogen: It is important to plant growth in the right quantity, it is one of the major ingredients in Fertilizers. Its properties also makes it useful in protecting valuable products from harmful contaminants.
2. Oxygen: Oxygen is needed by living things to survive. During respiration, animals take in oxygen which is used to breakdown food and produce energy.
3. Carbon dioxide: It is used by plants during photosynthesis to manufacture their food.
4. Water vapour: It condenses in the atmosphere to form clouds. It also helps to preserve heat within the earth's atmosphere.
5. Dust particles: Dust is an air pollutant but it helps provide a surface for water vapour in the atmosphere to condense and form clouds leading to precipitation (rainfall).